

Lloyd Falltrick

<https://portfolio-beta-taupe-iomiubi5kw.vercel.app/>

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EDUCATION

University of Brighton

Bachelors of Science in Computer Science

Brighton, UK

Sept. 2024 – Jun. 2027

EXPERIENCE

Outthink

Software Engineer Fellow

Brighton, UK

Jan 2026 - May 2026

- **React/TypeScript:** Implemented reusable React/TypeScript components and accessibility improvements across a security-awareness platform serving 500,000+ enterprise users, reducing UI inconsistencies by standardising design tokens and shared layouts.
- **Figma MCP:** Worked directly in figma with designers - utilising the MCP server to implement multiple design changes or new features with breakneck pace.
- **Python:** Developed Python automation scripts wrapping FFmpeg CLI workflows, cutting multi-step video processing tasks from minutes to seconds and eliminating repetitive manual commands.

Specsavers

Optical Assistant

UK

Sep 2022 - Dec 2025

- **Troubleshooting:** Troubleshooted technical issues with glasses and hearing aids, diagnosing faults and explaining solutions to customers.

PROJECTS

3D SPA | *React, Typescript, Three.js:*

3D portfolio using three.js and react typescript imitating a 3D computer, with an operating system including games and projects.

- Built a single-page application integrating 3D elements while preserving DOM structure and accessibility, with support for interactions on low-end machines overcoming typical canvas-based limitations using Three.js and CSS3D.
- Delivered a technical talk to the Brighton web community on accessible 3D web design patterns.

Unity/Unreal Game prototypes | *Unity, C#, HLSL:*

Dozens of prototypes to pushing the engine and find out what is possible, whether shader, scripting, or engine work.

- Created multiple prototypes exploring procedural generation, Gerstner waves, and interactivity, as well as other game systems.

Render Raycast 3D engine | *Java:*

A from-scratch pseudo-3D engine inspired by Wolfenstein 3D (1992).

- Implemented a raycasting engine inspired by Wolfenstein to deepen understanding of Java and low-level rendering concepts without using OpenGL or any 3D/game related libraries.

PROGRAMMING SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, Java, C#, SQL, HTML/CSS, HLSL

Technologies: React, Three.js, Unity, Git, FFmpeg, MCP, Scikit, Numpy

Concepts: OOP, Agile/Scrum, accessibility, 3D rendering, procedural generation, REST APIs, Machine Learning/Artificial Intelligence

Tools: GitHub, VS Code, Figma, Unity Editor

Interests: Drumming, Chess, Music, Games Programming, 3D modelling